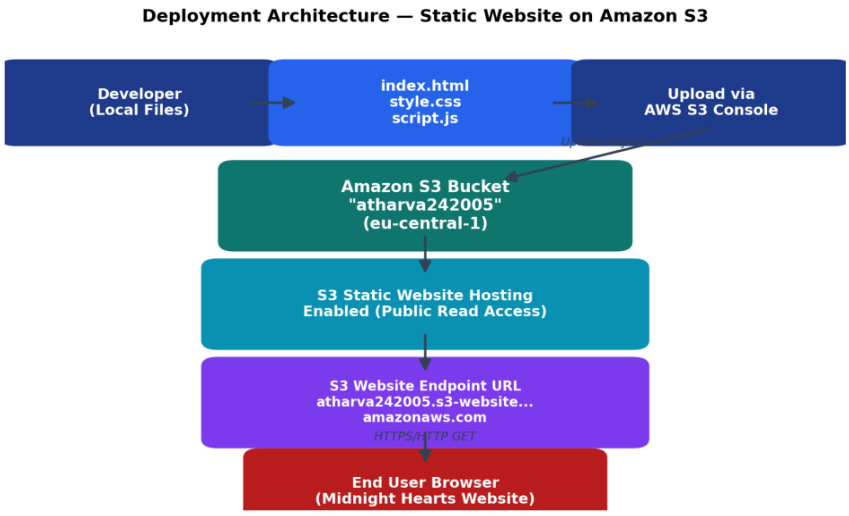
PROJECT REPORT

Hosting a Static Website Using Amazon S3

"Midnight Hearts" – A Mood-Based Romantic Quote Website



Cloud Platform: Amazon Web Services (AWS)

Service Used: Amazon S3 (Simple Storage Service)

Bucket Name: atharva242005 | Region: eu-central-1 (Frankfurt)

Date: September 2026

NAME : ATHARVA VADAGE

1. Abstract

This project demonstrates the deployment of a fully static website on the cloud using Amazon Web Services (AWS), specifically Amazon Simple Storage Service (S3). The website, named “Midnight Hearts,” is a mood-based romantic quote generator built using plain HTML, CSS, and JavaScript. Rather than relying on a traditional web server, the project leverages S3's static website hosting feature to serve the site directly from cloud storage, making it a low-cost, highly available, and easily scalable solution suitable for personal projects, portfolios, and lightweight web applications.

2. Objectives

- To design and develop a responsive front-end website using HTML, CSS, and JavaScript.
- To understand and implement AWS S3 bucket creation and configuration.
- To enable and configure static website hosting on an S3 bucket.
- To deploy website assets (HTML, CSS, JS) to the cloud without a dedicated backend server.
- To test public accessibility of the deployed website through the S3 website endpoint.

3. Technology Stack

Layer	Technology Used
Frontend Structure	HTML5 (index.html)
Styling	CSS3 (style.css)
Interactivity / Logic	JavaScript (script.js)
Cloud Hosting	Amazon S3 (Static Website Hosting)
Cloud Region	Europe – Frankfurt (eu-central-1)
Access Control	Public read access via bucket policy

4. System / Deployment Architecture

The architecture follows a simple, serverless static-hosting model. Local project files are uploaded directly to an S3 bucket, which is configured to serve them as a public website. The end user's browser communicates directly with the S3 website endpoint over HTTP/HTTPS, with no application server or database involved.

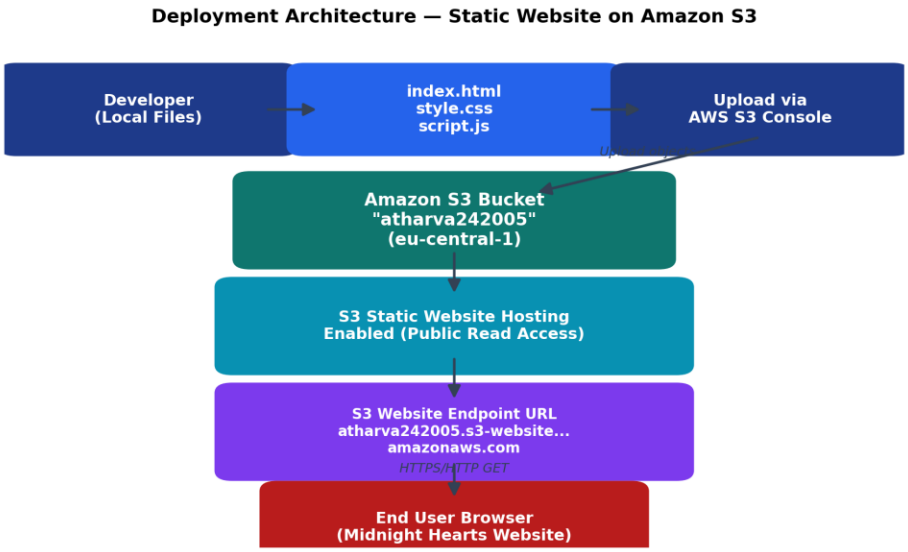


Figure 1: Deployment architecture of the static website on Amazon S3

5. Implementation Steps

The following steps were carried out to deploy the website:

- Created an S3 bucket named “atharva242005” in the eu-central-1 (Frankfurt) region.
- Uploaded the three website files — index.html, script.js, and style.css — as objects into the bucket.
- Enabled “Static website hosting” under the bucket's Properties tab and set index.html as the index document.
- Configured a bucket policy / permissions to allow public read (GetObject) access to the objects.
- Retrieved the S3 website endpoint URL and verified the site loads correctly in a browser.
- Tested navigation, interactivity (mood selection, quote generation, favoriting), and page responsiveness.

6. Application Features

- Landing page with a call-to-action button (“Discover Your Quote”).
- Mood-based quote filtering — users select a mood (e.g., Happy, Sad) to receive a matching quote.
- “Another Quote” button to fetch a new random quote for the same mood.
- “Favorite” button to save a preferred quote.
- Navigation bar with Home, Your Mood, Gallery, and About sections.
- Fully responsive, dark-themed romantic UI built with custom CSS.

7. Application / User Flow

The diagram below illustrates the typical interaction flow a visitor follows when using the Midnight Hearts website, from landing on the home page to receiving and optionally saving a quote.

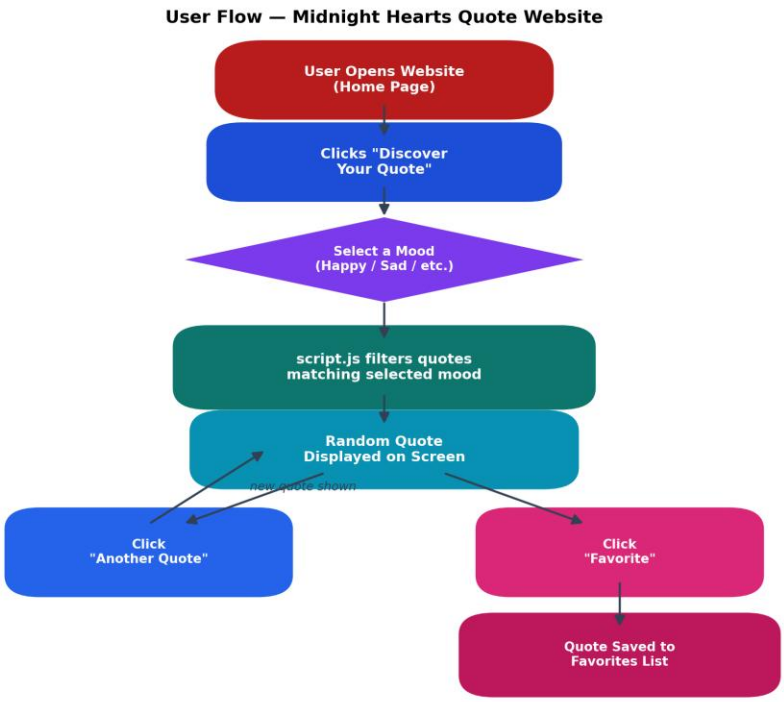


Figure 2: User flow within the Midnight Hearts application

8. Screenshots

Amazon S3 bucket showing the three uploaded website objects (index.html, script.js, style.css):

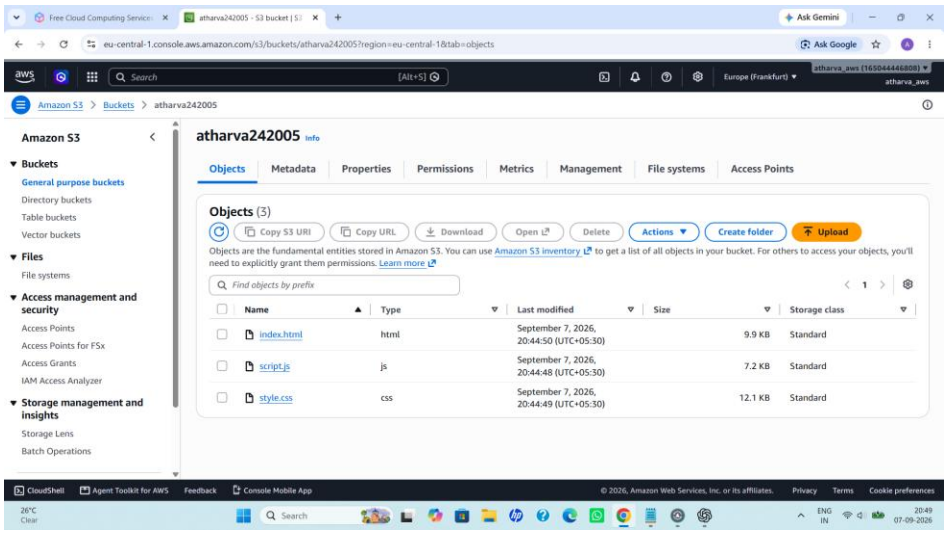


Figure 3: S3 bucket "atharva242005" — Objects view

Home page of the deployed "Midnight Hearts" website:

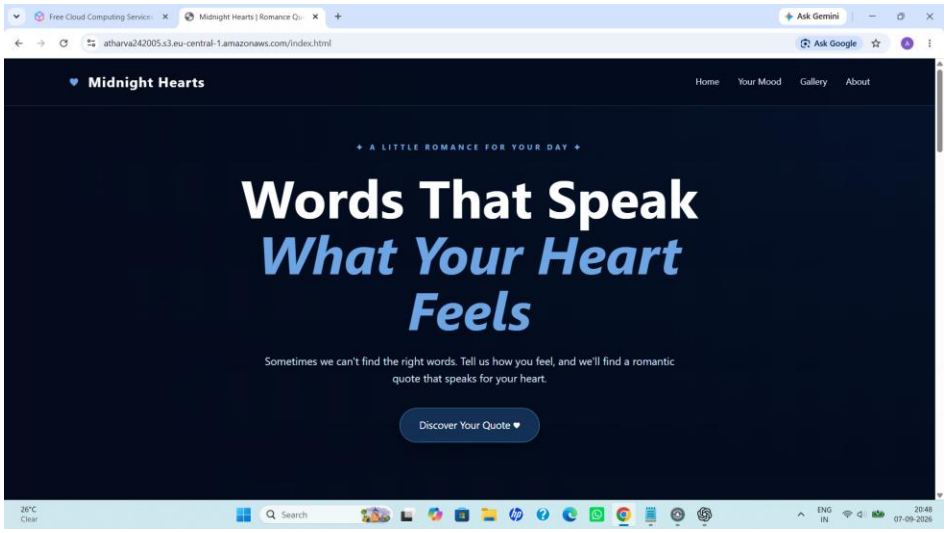


Figure 4: Website home page served from the S3 endpoint

Quote result page after selecting a mood:

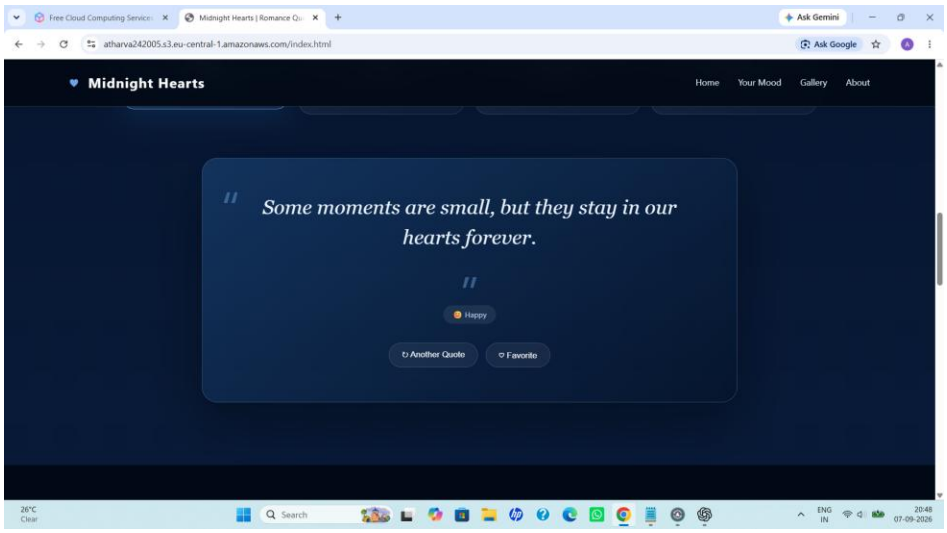


Figure 5: Mood-based quote result screen

9. Challenges Faced

- Ensuring correct MIME types were served for .html, .css, and .js objects.
- Configuring bucket permissions carefully to allow public access only to required objects.
- Verifying that index.html was correctly set as the default document for the website endpoint.

10. Conclusion

The project successfully demonstrates how a static website can be hosted on the cloud using Amazon S3 without the need for a traditional web server or backend infrastructure. This approach is cost-effective, scalable, and easy to maintain, making it ideal for portfolios, landing pages, and small interactive front-end applications such as the "Midnight Hearts" quote website built in this project.

11. Future Scope

- Integrate Amazon CloudFront as a CDN for faster global content delivery and HTTPS support.
- Use Route 53 to map the S3 website to a custom domain name.
- Add a backend (e.g., AWS Lambda + DynamoDB) to persist favorited quotes per user.
- Automate deployment using AWS CLI or a CI/CD pipeline (e.g., GitHub Actions).